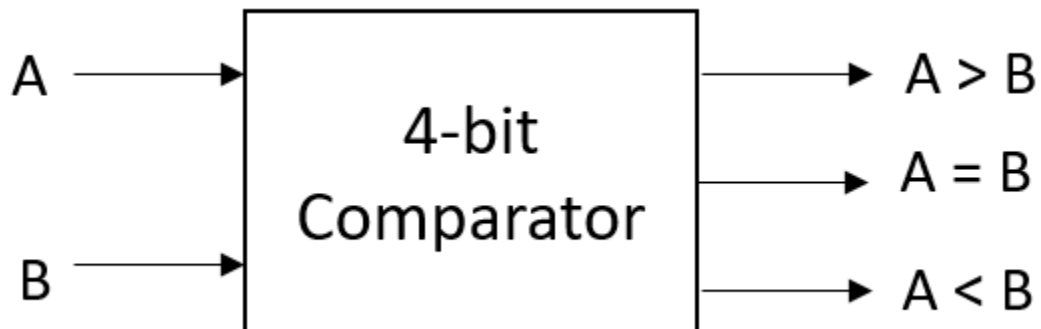


Day 18

4-bit Comparator in Verilog

Circuit Diagram :



Verilog Code

4-bit comparator code

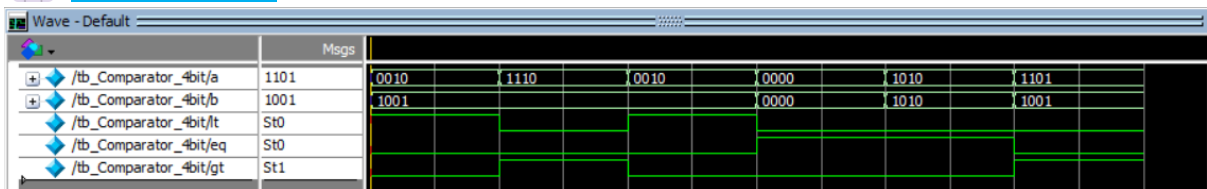
```
1 module Comparator_4bit(a,b,lt,eq,gt );
2 input [3:0] a,b;
3 output lt,eq,gt;
4 assign lt=(a<b);
5 assign eq=(a==b);
6 assign gt=(a>b);
7 endmodule
8
```

Testbench Code

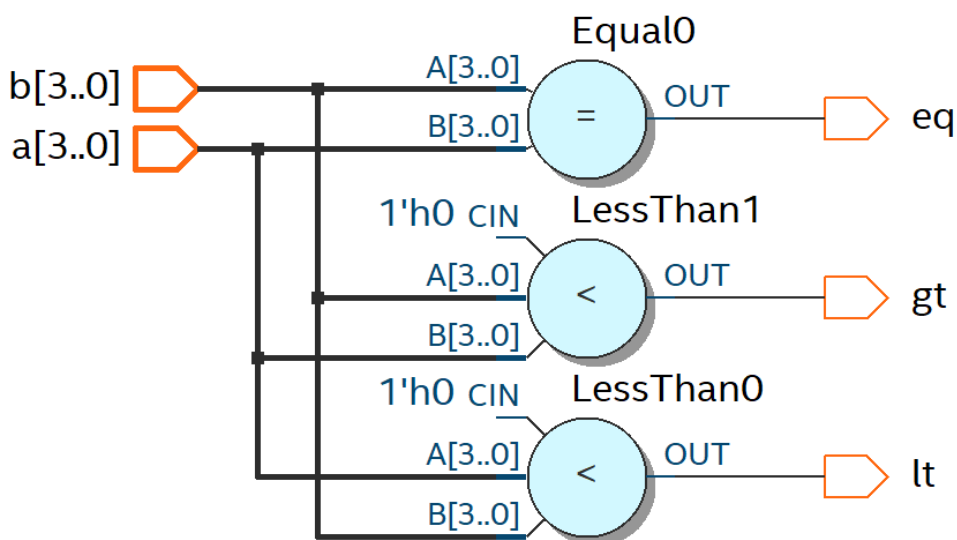
4-bit comparator Tb code

```
1 module tb_Comparator_4bit;
2   reg [3:0] a,b;
3   wire lt,eq,gt;
4   Comparator_4bit dut(a,b,lt,eq,gt);
5   initial
6   begin
7     $monitor("Time=0t a=%b b=%b lt=%b eq=%b gt=%b", $time,a,b,lt,eq,gt);
8     a=4'b0010; b=4'b1001; #10
9     a=4'b1110; b=4'b1001; #10
10    a=4'b0010; b=4'b1001; #10
11    a=4'b0000; b=4'b0000; #10
12    a=4'b1010; b=4'b1010; #10
13    a=4'b1101; b=4'b1001; #10
14    $finish;
15  end
16 endmodule
```

Waveform



Schematic





EDA Tools Used

- IntelQuartusPrime
- ModelSim

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